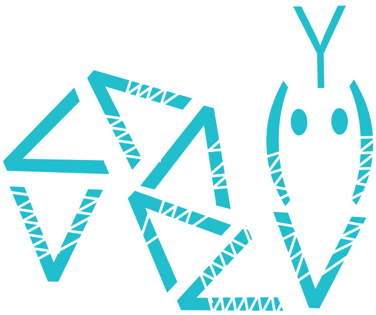
AI: Anaconda and More on Probability

In the previous article in this series on AI and machine learning, we began by exploring the intricacies of using TensorFlow, a very powerful library used for developing AI and machine learning applications. Then, we discussed probability and paved the way for many of our future discussions. In this article, the fifth in the series, we will continue exploring concepts in probability and statistics.



To begin with, let us first install Anaconda, a distribution of Python for scientific computing especially useful for developing AI, machine learning and data science based applications. Later, we will also introduce a Python library called Theano in the context of probability theory. But before doing that, let us discuss the future of AI a little bit.

While going through the earlier articles in this series for evaluation and course correction, I wondered whether my tone was a bit sceptical at times while discussing the future of AI. By being brutally honest about some of the topics we discussed, I may have inadvertently demotivated a tiny fraction of the readers and potential practitioners of AI. This made me research the financial aspects of AI and machine learning. I wanted to identify the kind of companies involved in the AI market. Are the bigwigs heavily involved or are there just some startups trying to move away from the pack? Another important question was: how much money will be pumped into the AI market by these companies in the near future? Is it going to be just a couple of million, or a few billion or maybe even a few trillion?

For the predictions and data stated here, I depended on articles published in respected newspapers in the recent past rather than on scholarly articles which I couldn't grasp well to understand the complex dynamics behind the development of an AI based economy. An article published by *Forbes* in had 2020 predicted that companies would spend 50 billion dollars on AI in financial year 2020. Well, that is a really huge investment, even if we acknowledge that the American billion (10^9) is just one-thousandth of the British billion (10^{12}). An article published in *Fortune*, another American business magazine, reported that venture capitalists are shifting some of their focus from artificial intelligence to newer and trendier fields like Web3 and decentralised finance (DeFi). But *The Wall Street Journal* (an American business-focused newspaper) in 2022 confidently predicted that, 'Big Tech Is Spending Billions on AI Research. Investors Should Keep an Eye Out'.

What about the Indian scenario? *Business Standard* reported in 2022 that 87 per cent of the Indian companies will hike their AI expenditure by 10 per cent in the next three years. So, overall, it looks like the future of AI is very safe and bright. But which are the top companies investing in AI? As expected, all the giants like Amazon, Meta (Facebook), Alphabet (Google), Microsoft, IBM, etc, are doing so. But, surprisingly, companies like Shell, Johnson & Johnson, Unilever, Walmart, etc, whose primary focus is neither computing nor information technology, are also in the list of big companies heavily investing in AI. And yes! Amazon is a tech company and not an e-commerce company.

So it is clear that many of the giant companies in the world believe that AI is going to play a prominent role in the near future. But what changes and new trends are these leaders expecting in the future? Again, to find some answers, I depended on news articles and interviews, rather than on scholarly articles. The terms frequently mentioned in context with future trends in AI include responsible AI, quantum AI, IoT with AI, AI and ethics, automated machine learning, etc. I believe these are vast topics and we will discuss some of these terms (except AI and ethics, which we already discussed in the last article) in detail, in the coming articles in this series.

An introduction to Anaconda

Let us now move to discussing the necessary tech for AI. From this article onwards, we will start using Anaconda whenever necessary. Anaconda is a distribution of the Python and R programming languages for scientific computing. It helps a lot by simplifying package management.